

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.4 MAGNETISM AND ELECTROMAGNETIC INDUCTION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the whole sub-strand. Answer the driving question: How does moving a wire through a magnetic field produce electricity, and how is this principle powering homes, farms and industries across Kenya? Use what you observed in our experiments, not just what you remember. Write in full sentences under each prompt. Use the words you learned, such as magnetic field lines, relative motion, induced e.m.f., galvanometer, coil turns and magnetic flux. Refer to the Kericho student, her bicycle dynamo and the school generator in your answers. Where a prompt asks for a diagram, draw and label it clearly. You may use your cumulative model and the Driving Question Board in your notes.

Part 1: Explaining the Kericho Puzzle

A Form One student in a boarding school in Kericho spins the wheel of a bicycle dynamo and a small LED lights up. When she holds the wheel still, the light goes out at once. There is no battery. (a) Explain why the LED lights up only when the wheel is spinning. (b) Explain what is happening to the magnetic field lines and the conductor while the light is on. (c) Use your galvanometer results from the electromagnetic induction experiment as evidence in your explanation.

(a) The LED lights only while the wheel spins because electromagnetic induction needs relative motion between a conductor and a magnetic field. Spinning the wheel makes the coil inside the dynamo move relative to the magnet. This induces an e.m.f. that pushes a current through the LED. When the wheel stops, there is no relative motion, so no e.m.f. is induced and the LED goes out. No battery or chemical reaction is involved. Mechanical motion is turned directly into electrical energy. (b) While the wheel spins, the conductor keeps cutting across the magnetic field lines, so the magnetic flux through the coil keeps changing. This continuous cutting sustains the induced e.m.f. and a steady current. (c) In our experiment, a copper conductor held still between the poles of a U-shaped magnet gave zero galvanometer deflection. Moving it parallel to the field lines gave zero or negligible deflection. Moving it upward across the field lines gave a clear deflection, and moving it at an angle gave an intermediate deflection. The needle also returned to zero as soon as movement stopped, just like the dynamo LED going out when the wheel stops.

Part 2: Why Faster Spinning Gives a Brighter Light

The Kericho student notices that when she spins the dynamo wheel faster, the LED

(a) The four factors are: (1) Speed of movement: pushing the magnet into the coil faster gave a larger deflection. (2) Strength of the magnetic field: two stacked magnets gave a larger deflection than one. (3) Number of coil turns: the 100-turn coil gave a larger deflection than the 20-turn coil. (4) Angle of movement: moving at 90° to the field gave the maximum deflection, and moving at 0°

becomes brighter. (a) Name the four factors that control the size of the induced e.m.f. and state, from your Lesson 5 observations, how each one changes the galvanometer deflection. (b) Using one of these factors, explain why faster spinning makes the LED brighter. (c) Explain how you made your Lesson 5 investigation a fair test.	(parallel) gave zero deflection. (b) Faster spinning makes the coil cut the magnetic field lines at a greater rate, so the magnetic flux changes more quickly. By Faraday's law, the induced e.m.f. is proportional to the rate of change of magnetic flux. A larger e.m.f. drives a larger current through the LED, so it shines brighter. (c) I changed only one factor at a time, for example the number of coil turns, and kept the others the same: the same galvanometer, the same magnet, the same speed and the same angle. I recorded the results in a table so I could compare them clearly.
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Part 3: Magnets, Magnetic Fields and Why Stillness Matters

(a) Describe what you learned about magnetic field lines around a U-shaped magnet, including their direction, where they are crowded, and what the field is like between the poles. Explain why this region is important in a generator. (b) Describe two ways you saw a magnet lose its magnetism in Lesson 2 and explain, using magnetic domains, why this happens. (c) Explain why engineers at Kenya Power protect generator magnets from heat and vibration.	(a) Outside the magnet, field lines run from the North pole to the South pole. They are most crowded at the poles, where the field is strongest. Between the poles of a U-shaped magnet the lines are parallel and evenly spaced, so the field is uniform and intense. This is the region where the coils of a generator rotate. The strong, constant field there gives the largest induced e.m.f. as the conductor cuts across the lines. (b) A magnetised nail heated red-hot over a jiko no longer deflected a compass needle. A magnetised iron rod hammered 20 times on an anvil block also lost its deflection. A third method used an AC solenoid. Magnetism depends on magnetic domains lined up in one direction. Heat and hammering give the domains enough energy to turn in random directions, so their effects cancel and the magnetism is destroyed. (c) If heat or vibration disordered the domains in a generator's magnets, the magnetic field would weaken or disappear. The coils would then have no strong field to cut, little or no e.m.f. would be induced, and there would be a power failure.
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Part 4: From a Bicycle Dynamo to Kenya's National Grid

(a) Explain why the same principle that lights a bicycle dynamo LED can also power the dormitory in Kericho and Kenya's national grid. (b) Choose two applications of electromagnetic induction from our research (for example AC generators, transformers, Olkaria Geothermal Plant, Lake	(a) All of them rely on relative motion between a conductor and a magnetic field, which induces an e.m.f. The school generator is the dynamo scaled up. Its large copper coils spin at high speed, have many turns, and sit in strong magnetic fields, so they produce enough e.m.f. for lights, water pumps and kitchen equipment. The same four factors apply at every scale. (b) Lake Turkana Wind Farm: wind turns the turbine blades, which spin generator coils relative to magnets. The relative motion induces an e.m.f. The faster the rotation, the greater the e.m.f. Olkaria Geothermal Plant: steam turns turbines that rotate coils in a magnetic field. Many turns of wire and strong electromagnets give a large e.m.f. Step-up transformers raise the voltage to 132 kV for transmission, and step-down transformers reduce it to 240 V for homes. Both rely on a changing magnetic flux in a coil. (c) When I closed the circuit, current flowed through the coil around the iron nail and magnetised it, making it an electromagnet. The nail attracted the metal strip (armature), which touched the nail and broke the circuit. With no current, the nail demagnetised and the strip sprang back, closing the circuit again. This repeated make-and-break action produced the ringing. More coil turns or more
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Turkana Wind Farm, electric bells, bicycle dynamos). For each, say how it uses induction and how it connects to the factors you studied. (c) Describe how the electric bell you built in Lesson 7 works, using the words magnetisation and demagnetisation.	current made the ring louder and faster.
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Part 5: Final Answer to the Driving Question

Write a final answer, in a connected paragraph of about 8 to 10 sentences, to the driving question: How does moving a wire through a magnetic field produce electricity, and how is this principle powering homes, farms and industries across Kenya? Include: the name of the principle, what must happen for an e.m.f. to be induced, what happens when movement stops, the factors that make the e.m.f. larger, and at least two Kenyan examples of how this helps daily life (this answers the KICD question: How is the study of electromagnetic induction important in our daily life?). You may also add a labelled diagram of a generator or dynamo.	The principle is electromagnetic induction. When a conductor such as a coil of copper wire moves so that it cuts across magnetic field lines, the magnetic flux through it changes and an e.m.f. is induced. If the circuit is complete, this e.m.f. drives a current. Relative motion between the conductor and the field is essential. When the movement stops, as when the Kericho student holds her bicycle dynamo still, the flux no longer changes, the induced e.m.f. falls to zero and the LED goes out. No battery or chemical reaction is needed, because mechanical energy is changed directly into electrical energy. The size of the e.m.f. increases with faster movement, a stronger magnetic field, more turns of wire, and movement at 90° to the field, which all follow from Faraday's law. Engineers use all of these in large generators, so the school generator in Kericho can light a dormitory and run water pumps and kitchen equipment. At the Olkaria Geothermal Plant and the Lake Turkana Wind Farm, turbines spin generator coils, and transformers step the voltage up to 132 kV for transmission and down to 240 V for homes. This electricity lights homes, runs water pumps on Rift Valley farms and powers machinery in industries such as fish processing in Kisumu. So the study of electromagnetic induction matters in daily life because it explains how Kenya's electricity is made and delivered.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining electromagnetic induction	Clearly explains that relative motion causes the conductor to cut field lines, changing flux and inducing an e.m.f. that drives current. Explains why stillness gives zero e.m.f. and says no battery is	States that movement between the conductor and the magnet induces an e.m.f. and that stopping stops the current, with limited explanation of field lines or flux.	Says only that spinning makes electricity or that the magnet makes it, without linking to relative motion or induced e.m.f.

	needed.		
Use of evidence from investigations	Uses specific results from the galvanometer experiments (stationary, parallel, upward, angled, speed, magnets, turns, 90° versus 0°) to support each claim in a Claim–Evidence–Reasoning way.	Uses some experimental results to support claims, but links to the reasoning may be incomplete.	Gives few or no observations from the experiments, or states results incorrectly.
Factors affecting induced e.m.f.	Names all four factors, correctly states how each changes the e.m.f., links them to Faraday's law and rate of change of flux, and explains fair testing.	Names at least three factors with mostly correct effects, with a limited link to Faraday's law.	Names one or two factors, or gives incorrect effects.
Magnetism concepts and applications in Kenya	Accurately describes field lines and domains, the bell's magnetisation and demagnetisation, and at least two Kenyan applications, including the role of transformers, using correct terms.	Describes most magnetism ideas and one or two applications, with small errors or missing detail.	Gives vague or incorrect descriptions and few or no Kenyan applications.
Communication and connection to the driving question	Writes a coherent, well-organised final answer that links the Kericho phenomenon to the national grid and daily life, using scientific vocabulary accurately.	Writes a mostly clear answer that addresses the driving question, with some vocabulary slips or missing links.	Writes a disorganised or incomplete answer that does not fully address the driving question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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